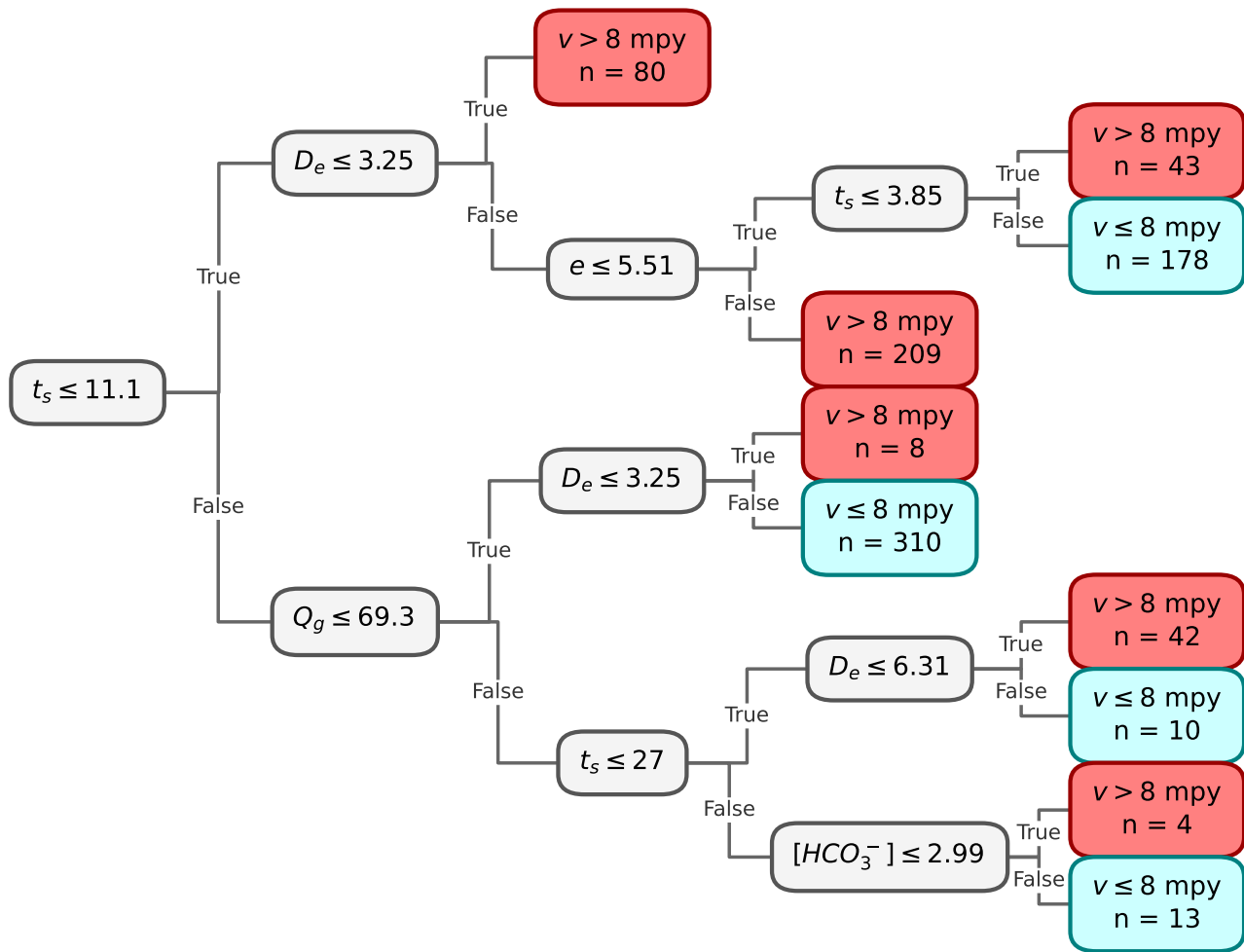




De: external pipe diameter [in]     $HCO_3^-$ : bicarbonate concentration [meq/L]     $Q_g$ : gas production rate [MMSCFD]  
e: wall thickness [mm]     $t_s$ : service time of the monitoring point [yr]